

Induced Maps

Lecture 8 - CMSE 890

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Goals

Goals for today:

- Filtrations
- Betti curves
- Induced Maps

Section 1

Filtrations

Simplicial Complex Filtration

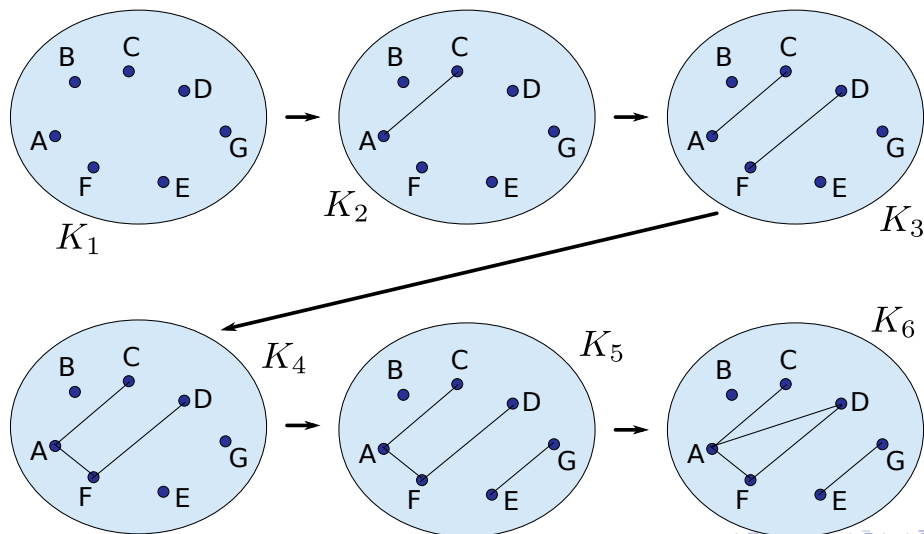
Definition

A filtration $\mathcal{F} = \mathcal{F}(K)$ of a simplicial complex K is a nested sequence of its subcomplexes

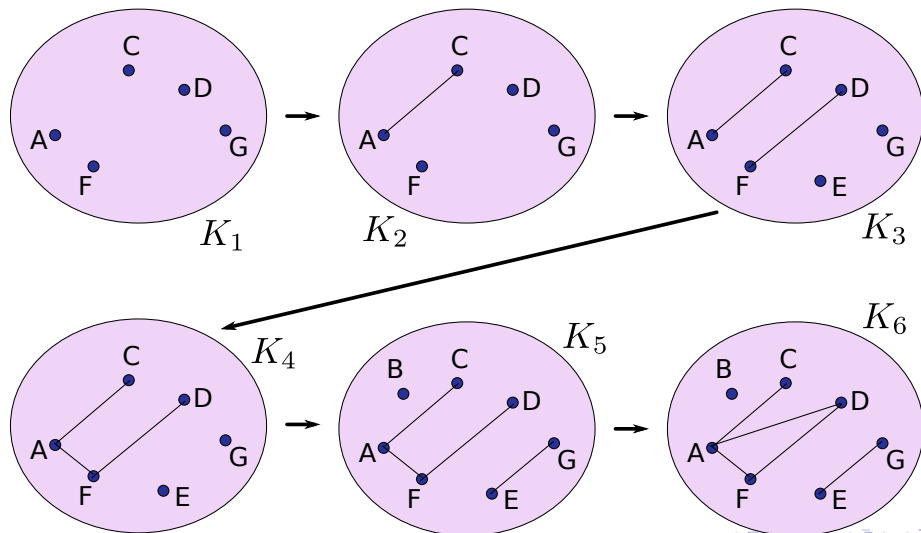
$$\mathcal{F} : \emptyset = K_0 \subseteq K_1 \subseteq K_2 \subseteq \cdots \subseteq K_n = K.$$

$\mathcal{F}$ is called *simplex-wise* if $K_i \setminus K_{i-1}$ is empty or a single simplex.

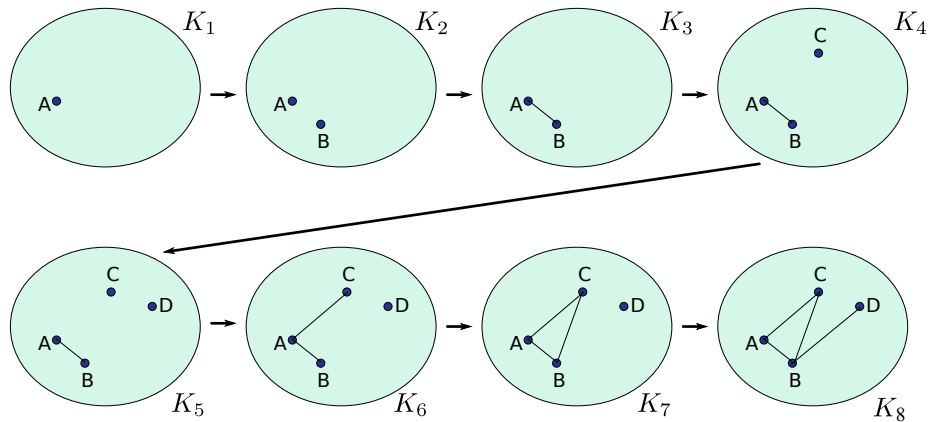
Example 1



Example 3



Example 3

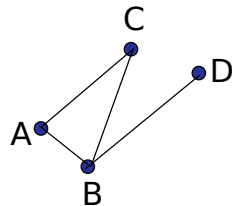
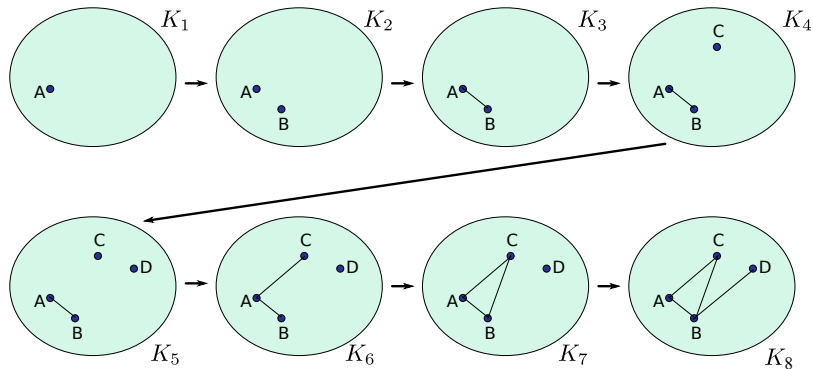


Simplex-wise monotone function

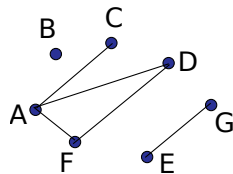
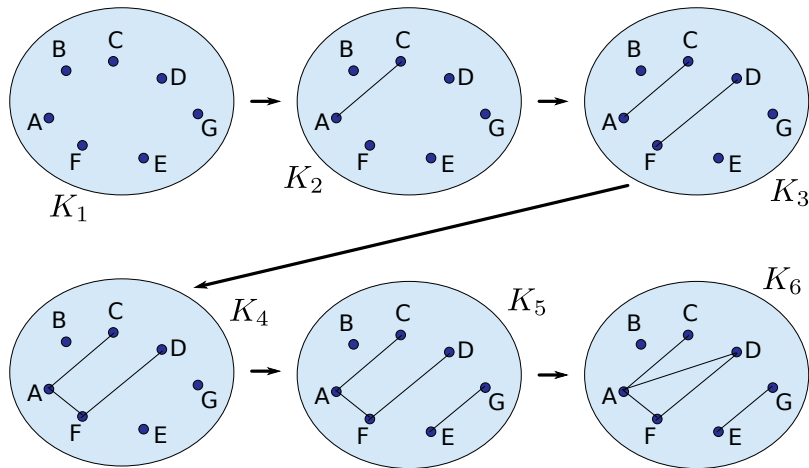
A simplex-wise function $f : K \rightarrow \mathbb{R}$ is called monotone if for every $\tau \leq \sigma$, $f(\tau) \leq f(\sigma)$. Fix values $a_0 \leq a_1 \leq \cdots a_n$. Let $K_i = f^{-1}(-\infty, a_i]$. The sublevelset filtration induced by this function is defined to be

$$\emptyset = K_0 \subseteq K_1 \subseteq K_2 \subseteq \cdots \subseteq K_n = K.$$

Example: Function inducing filtration

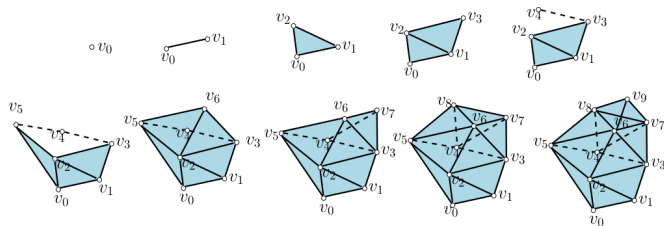


Tryit: Function inducing filtration



Everything comes from a function

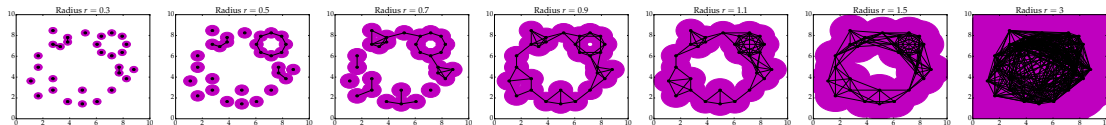
Vertex valued function (lower star filtration)



- Given $f : V \rightarrow \mathbb{R}$ defined on vertices of K .
- Extend to simplices by $f(\sigma) = \max_{v \in \sigma} f(v)$.

Everything comes from a function

Rips complex filtration



Section 2

What to do with a filtration: Betti Curves

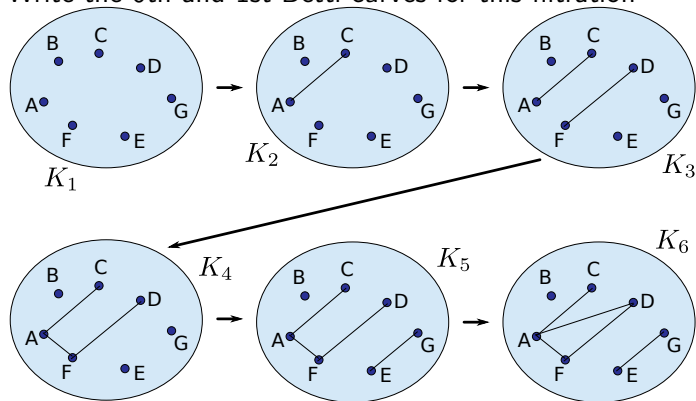
Betti curve

The p -th Betti curve is a function

$$\begin{aligned}\beta_p(\mathcal{F}) : \mathbb{Z} &\rightarrow \mathbb{Z} \\ i &\mapsto \beta_p(K_i) = \text{rk}(H_p(K_i))\end{aligned}$$

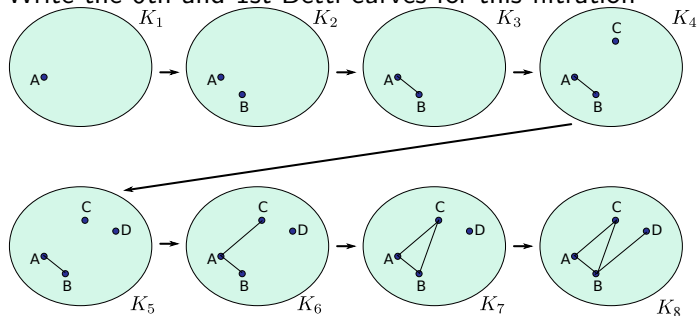
Example: Betti curves

Write the 0th and 1st Betti curves for this filtration



Tryit: Betti curves

Write the 0th and 1st Betti curves for this filtration



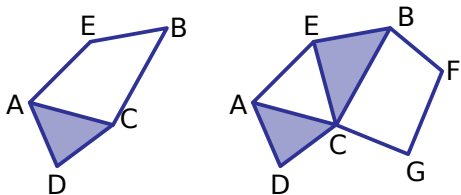
Section 3

Induced maps

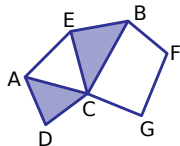
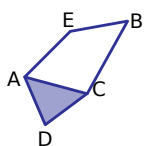
Simplicial maps

A simplicial map between abstract simplicial complexes $f : K \rightarrow L$ is induced by a map on the vertices $f : V(K) \rightarrow V(L)$.

(In this class, we care about inclusions.....)



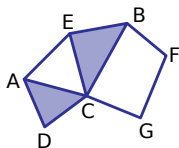
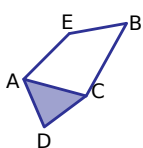
Big diagrams of vector spaces, $f_{\#}$



$$\cdots \longrightarrow C_2(K) \xrightarrow{\partial_2^K} C_1(K) \xrightarrow{\partial_1^K} C_0(K) \xrightarrow{\partial_0^K} 0$$

$$\cdots \longrightarrow C_2(L) \xrightarrow{\partial_2^L} C_1(L) \xrightarrow{\partial_1^L} C_0(L) \xrightarrow{\partial_0^K} 0$$

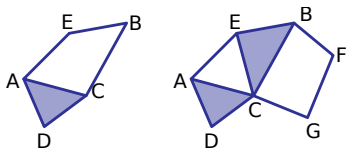
Induced map



Given a simplicial map $f : K \rightarrow L$, the induced map on homology is defined by

$$\begin{aligned} f_* : H_p(K) &\rightarrow H_p(L) \\ [\alpha] &\mapsto [f_{\#}(\alpha)] \end{aligned}$$

Commutative diagram



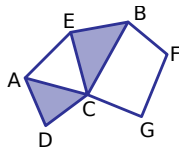
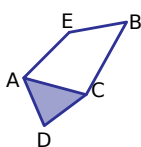
A diagram commutes if any path of functions are equal.

$$\begin{array}{ccccccc}
 \cdots & \longrightarrow & C_2(K) & \xrightarrow{\partial_2^K} & C_1(K) & \xrightarrow{\partial_1^K} & C_0(K) \xrightarrow{\partial_0^K} 0 \\
 & & \downarrow f_{\#} & & \downarrow f_{\#} & & \downarrow f_{\#} \\
 \cdots & \longrightarrow & C_2(L) & \xrightarrow{\partial_2^L} & C_1(L) & \xrightarrow{\partial_1^L} & C_0(L) \xrightarrow{\partial_0^K} 0
 \end{array}$$

Claim: $f_{\#} \circ \partial^K = \partial^L \circ f_{\#}$.

Effect on cycles

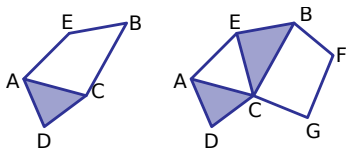
Cycles go to cycles!



$$\begin{array}{ccccccc}
 \dots & \longrightarrow & C_2(K) & \xrightarrow{\partial_2^K} & C_1(K) & \xrightarrow{\partial_1^K} & C_0(K) \xrightarrow{\partial_0^K} 0 \\
 & & \downarrow f_{\#} & & \downarrow f_{\#} & & \downarrow f_{\#} \\
 \dots & \longrightarrow & C_2(L) & \xrightarrow{\partial_2^L} & C_1(L) & \xrightarrow{\partial_1^L} & C_0(L) \xrightarrow{\partial_0^K} 0
 \end{array}$$

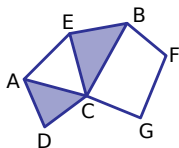
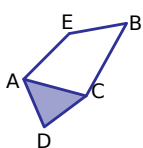
Effect on boundaries

Boundaries go to boundaries!



$$\begin{array}{ccccccc}
 \cdots & \longrightarrow & C_2(K) & \xrightarrow{\partial_2^K} & C_1(K) & \xrightarrow{\partial_1^K} & C_0(K) \xrightarrow{\partial_0^K} 0 \\
 & & \downarrow f_{\#} & & \downarrow f_{\#} & & \downarrow f_{\#} \\
 \cdots & \longrightarrow & C_2(L) & \xrightarrow{\partial_2^L} & C_1(L) & \xrightarrow{\partial_1^L} & C_0(L) \xrightarrow{\partial_0^K} 0
 \end{array}$$

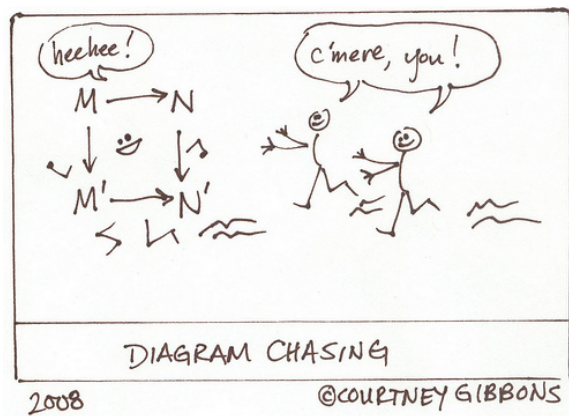
Repeat: Induced map



Given a simplicial map $f : K \rightarrow L$, the induced map on homology is defined by

$$\begin{aligned} f_* : H_p(K) &\rightarrow H_p(L) \\ [\alpha] &\mapsto [f_{\#}(\alpha)] \end{aligned}$$

Diagram Chasing



Try it:

If $H_1(K)$ generated by

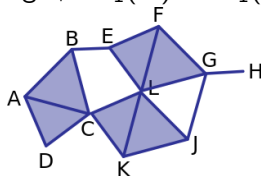
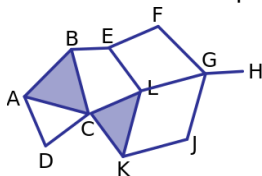
$$[\alpha_1] = [AC + CD + AD]$$

$$[\alpha_2] = [BE + EL + CL + CB]$$

$$[\alpha_3] = [EF + FG + GL + EL]$$

$$[\alpha_4] = [GL + GJ + JK + LK]$$

write the matrix representing $f_* : H_1(K) \rightarrow H_1(L)$ induced by inclusion.

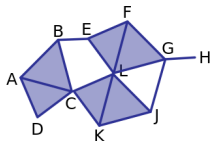
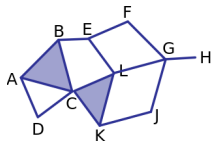


and $H_1(L)$ generated by

$$[\beta_1] = [BE + EL + CL + CB]$$

$$[\beta_2] = [GL + GJ + LJ]$$

More spare paper



$$\begin{array}{ccccccc}
 \cdots & \longrightarrow & C_2(K) & \xrightarrow{\partial_2^K} & C_1(K) & \xrightarrow{\partial_1^K} & C_0(K) \xrightarrow{\partial_0^K} 0 \\
 & & \downarrow f_{\#} & & \downarrow f_{\#} & & \downarrow f_{\#} \\
 \cdots & \longrightarrow & C_2(L) & \xrightarrow{\partial_2^L} & C_1(L) & \xrightarrow{\partial_1^L} & C_0(L) \xrightarrow{\partial_0^K} 0
 \end{array}$$

Next time: Persistence module

Given a filtration $\mathcal{F}$, the p -dimensional persistence module is

$$H_p(\mathcal{F}) : 0 = H_p(K_0) \rightarrow H_p(K_1) \rightarrow H_p(K_2) \cdots \rightarrow H_p(K_n) = H_p(K)$$

with maps induced by inclusion.

For next time

- No class 9/23 or 9/25
- Everyone homework:
 - ▶ Find the AATRN youtube channel
 - ▶ Find one of the Tutorial-a-thon playlists
 - ▶ Watch one video (or more, they're largely less than 15 minutes long). Write a few sentences about what you learned and any questions you have and post it with a link to the video on our slack channel.
 - ▶ Comment on at least one other person's post.